

MGTA 464: SQL and ETL

STUDENT CLASS (2027)

TERM (SU26)

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DESCRIPTION

Data is one of the most valuable assets in modern organizations, but its value depends on the ability to efficiently access, transform, and analyze it. This course provides students with the foundational and practical skills needed to work with data in real-world business environments through Structured Query Language (SQL) and Extract, Transform, and Load (ETL) processes. SQL remains the industry standard for interacting with relational databases and is one of the most sought-after technical skills in analytics, consulting, finance, marketing, operations, technology, and data science roles.

Students will learn how organizations store, manage, and retrieve data from enterprise databases, and how to write effective SQL queries to answer business questions, generate insights, and support data-driven decision-making. Beyond querying data, the course introduces ETL concepts and workflows that enable analysts to collect data from multiple sources, clean and transform it, and prepare it for reporting, visualization, and advanced analytics. These skills form the backbone of modern business intelligence and analytics systems.

As organizations increasingly rely on large-scale data infrastructure, business analytics professionals are expected not only to interpret results but also to understand where data originates, how it is structured, and how it moves through analytical pipelines. Mastery of SQL and ETL empowers students to work more independently, communicate effectively with data engineers and technical teams, ensure data quality, and unlock insights that might otherwise remain inaccessible.

By the end of the course, students will be able to design and execute SQL queries of increasing complexity, integrate and transform data from diverse sources, and build reproducible data workflows that support business analysis. These capabilities are essential for success in graduate-level analytics coursework and are highly valued by employers seeking professionals who can bridge the gap between business decision-making and modern data infrastructure.

OBJECTIVES

- Understand and apply foundational database concepts to organize, manage, and access business data effectively.
- Retrieve and manipulate data using SQL to answer business questions, generate insights, and support data-driven decision-making.
- Apply advanced SQL techniques, including joins, subqueries, aggregations, and common table expressions, to analyze complex datasets and solve analytical problems.
- Design and implement ETL workflows that extract, transform, and load data from multiple sources into structured, analysis-ready datasets.

MATERIALS

All materials will be provided to you including a mix of powerpoint slides, data sets, and online reference material including books.

COURSE SCHEDULE

Session	Topic and Activity
1	Course introduction Working with databases SQL Module A
2	Quiz #1 SQL Module B
3	Quiz #2 SQL Module C
4	Quiz #3 SQL Module D ETL Module A
5	Quiz #4 ETL Module B Final Project (due 9/3 at 8am)

DELIVERABLES

In-class Quizzes

At the beginning of sessions 2 through 5, there will be an in-class quiz. The quizzes are conducted in Respondus Lockdown Browser. Be sure to have this browser installed and working well on your laptop as technical problems tend to eat into your quiz time. A sample lockdown browser quiz will be made available so that you can test your environment. Each quiz will test your understanding of the material covered in the sessions and assignments prior to the quiz. By carefully working through the lecture material and assignments, you will be able to keep up with the class and perform better on these quizzes.

Final Project

This course has a final project as the last deliverable. This project will require you to leverage everything you learned in the course as part of an ETL case.

Class Participation

Your class participation grade is based upon your presence and active learning in class (10%) and your contribution to in-class exercises (10%).

GRADING

Deliverables	Percentage
In-class quizzes	70%
Final project	10%
Class Participation	20%
Total	100%

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

ACADEMIC INTEGRITY VIOLATIONS

If you are found responsible for an academic integrity violation after going through the Academic Integrity Office's (AIO) process, then in addition to the administrative sanctions you receive from the university (<https://academicintegrity.ucsd.edu/process/consequences/sanctioning-guidelines.html>), you will receive academic consequences as well in the course. I strictly follow the faculty guidance on assessing academic consequences which can be found here: https://academicintegrity.ucsd.edu/files/process-files/Faculty-to-Faculty%20Advice%20on%20Academic%20Sanctions_Handout_PDF.pdf. Be sure to note that "Exam Cheating, Falsification/Fabrication/Fraud or Contract Cheating" will result in your receiving an "F" in the course, and thus you will be required to retake the course next year. The academic consequences are non-negotiable and do not depend on your particular circumstances (e.g., concerns about your fellowship, a company sponsoring you in the program, graduating on time, etc.). While I appreciate these are very real concerns, it is also easy to avoid by maintaining your academic integrity. It is much better to receive an average or poor grade, then to receive an "F", delay your graduation, and even possibly being removed from the program. I strongly encourage you to make smart academic integrity decisions in this course as well as your entire time here at UC San Diego.

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

CLASS POLICY ON THE USE OF ARTIFICIAL INTELLIGENCE TOOLS

The use of artificial intelligence (AI) tools such as ChatGPT, Bing Chat, Bard, Claude, etc. is encouraged in this class. These tools have considerable potential to streamline and augment our work. You should use them with the following caveats and conditions:

- AI-tools should enhance your work, not replace it. For instance, large language models (LLMs) can improve the quality of your writing, identify new sources, provide high-level summaries of complex ideas or terms, generate code, and more. These tools can help you come up with new ideas, overcome writer's block, and refine your original ideas. You should not use Generative AI tools to generate content that you simply put your name on. This constitutes an academic integrity violation, much like taking another individual's work and misrepresenting it as your own.
- You may utilize Generative AI and LLMs for class preparation and for help with assignments. However, we require that you fully understand and can explain, in-person or on video, all details of every line of code and mathematical calculation that you submit as your own work.
- If you use AI for an assignment, be transparent about how it was used in a footnote or citation (e.g., a link to the ChatGPT chat that you created).
- Be aware that LLMs have documented issues with hallucination and inaccuracy (e.g., generating fake python packages, failing logical arguments, and misrepresenting sources). You are fully responsible for all content you submit in your assignments and will be graded accordingly.